

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q22-2213

Anatolia 138 kV
September 21, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2213. The Interconnection Customer proposes a 80.1 MW large load (data center) facility interconnecting at the Anatolia 138 kV substation in the Maryland area. The request was received on October 23, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 479 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,877,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: HOOKDALE to COLUMBA 138.0 kV #2
Contingency: COLUMBIA to HOOKDALE 138.0 kV Cir. 1
Loading: 133.0% (Rating: 364 MVA)
DFAX: 5.5%
Pre-project loading: 131.8%
2. Facility: SIERRA PACIFIC LUMBER to CALIFORNIA AVE 138.0 kV
Contingency: LAKE CREEK to SIERRA PACIFIC LUMBER 138.0 kV
Loading: 131.4% (Rating: 233 MVA)
DFAX: 7.1%
Pre-project loading: 129.0%
3. Facility: FOOTHILL to WAUGH CHAPEL 138.0 kV line
Contingency: SIERRA PACIFIC LUMBER 138 kV/1 kV Transformer
Loading: 129.2% (Rating: 190 MVA)
DFAX: 8.5%
Pre-project loading: 125.6%
4. Facility: HOOKDALE to COLUMBIA 138.0 kV #1
Contingency: COLUMBIA to HOOKDALE 138.0 kV Line 2
Loading: 124.6% (Rating: 389 MVA)
DFAX: 5.5%
Pre-project loading: 123.5%
5. Facility: SANDARIO to GLEN BURNIE 138.0 kV
Contingency: LKE CRK to SIERRA PACIFIC LUMBER 138.0 kV L
Loading: 123.4% (Rating: 305 MVA)
DFAX: 18.1%
Pre-project loading: 118.7%

6. Facility: CHANNELVIEW SUBSTATION to CALIFORNIA AVE 138.0 kV L
Contingency: LAKE CREEK to SIERRA PACIFIC LUMBER 138.0 kV
Loading: 123.1% (Rating: 220 MVA)
DFAX: 7.1%
Pre-project loading: 120.5%
7. Facility: MCCALL SUBSTATION to FOOTHILL 138.0 kV L
Contingency: SIERRA PACIFIC LUMBER 138/1kV Transformer
Loading: 121.4% (Rating: 235 MVA)
DFAX: 8.5%
Pre-project loading: 118.5%
8. Facility: FOX RIV SUBSTATION to SANDARIO 138.0 kV L
Contingency: LAKE CREEK to SIERRA PACIFIC LUMBER 138.0 kV
Loading: 120.4% (Rating: 171 MVA)
DFAX: 7.1%
Pre-project loading: 117.1%
9. Facility: CHANNELVIEW SUBSTATION to GEORGIA-PACIFIC CROSSETT LLC SUBSTATION 138.0 kV
Contingency: LAKE CRK to SIERRA PACIFIC LUMBER SUBSTATION 138.0 kV
Loading: 116.8% (Rating: 206 MVA)
DFAX: 7.1%
Pre-project loading: 114.0%
10. Facility: HUNT CLUB SUBSTATION to SIERRA PACIFIC LUMBER 138.0 kV
Contingency: LAKE CREEK to SIERRA PACIFIC LUMBER 138.0 kV
Loading: 116.7% (Rating: 232 MVA)
DFAX: 5.9%
Pre-project loading: 114.6%
11. Facility: PORT ROSON to HOLOPAW 138.0 kV
Contingency: TABER to PEBBLEDAL 138.0 kV
Loading: 115.4% (Rating: 387 MVA)
DFAX: 15.2%
Pre-project loading: 112.2%
12. Facility: GEORGIA-PACIFIC CROSSETT LLC SUBSTATION to FOX RIVER 138.0 kV
Contingency: LAKE CRK to SIERRA PACIFIC LUMBER SUBSTATION 138.0 kV
Loading: 115.0% (Rating: 200 MVA)
DFAX: 7.1%
Pre-project loading: 112.1%
13. Facility: SANDARIO to STILLWATER HYDRO ELECTRIC PROJECT 138.0 kV
Contingency: SIERRA PACIFIC LUMBER 138 kV/1 kV Transformer Bank
Loading: 113.4% (Rating: 343 MVA)
DFAX: 15.5%
Pre-project loading: 109.8%
14. Facility: HUNT CLUB to SALAMANCAFRANK ST 138.0 kV Line
Contingency: SIERRA PACIFIC LUMBER to LAKE CRK 138.0 kV
Loading: 112.9% (Rating: 225 MVA)
DFAX: 5.9%
Pre-project loading: 110.8%
15. Facility: MCCALL SUBSTATION to STILLWATER HYDRO ELECTRIC PROJECT 138.0 kV
Contingency: SIERRA PACIFIC LUMBER 138/1kV TX
Loading: 112.5% (Rating: 364 MVA)
DFAX: 15.5%
Pre-project loading: 109.1%
16. Facility: FRENCH GULCH to TERRY SUBSTATION 138.0 kV
Contingency: SAN LORENZO 3302 to PAJUIL TC 138.0 kV Line

- Loading: 111.9% (Rating: 225 MVA)
DFAX: 7.0%
Pre-project loading: 109.4%
17. Facility: ALLIANCE to RIO BLANCO 2702 138.0 kV
Contingency: SIERRA PACIFIC LUMBER 138/1kV Transformer Bank
Loading: 111.4% (Rating: 404 MVA)
DFAX: 21.4%
Pre-project loading: 107.2%
18. Facility: BRUCE CROSSING to RIO BLANCO 2702 138.0 kV
Contingency: SIERRA PACIFIC LUMBER 138/1 kV TX
Loading: 110.5% (Rating: 433 MVA)
DFAX: 21.4%
Pre-project loading: 106.6%
19. Facility: BRUCE CROSSING to HOLOPAW 138.0 kV
Contingency: SIERRA PACIFIC LUMBER 138-1 kV Transformer Bank
Loading: 109.7% (Rating: 459 MVA)
DFAX: 21.4%
Pre-project loading: 106.0%
20. Facility: RALPH to FRENCH GUCH SUBSTATION 138.0 kV
Contingency: PAJUUL TC to SAN LORENZO 3302 138.0 kV Line
Loading: 107.7% (Rating: 219 MVA)
DFAX: 7.0%
Pre-project loading: 105.1%
21. Facility: COLUMBIA to SAN LORENZO 3302 SUBSTATION 138.0 kV
Contingency: TERRY to HOKDALE 138.0 kV
Loading: 107.3% (Rating: 218 MVA)
DFAX: 5.7%
Pre-project loading: 105.2%
22. Facility: TERRY to HOOKDALE 138.0 kV
Contingency: SAN LORENZO 3302 to PAJUUL TC 138.0 kV L
Loading: 104.3% (Rating: 285 MVA)
DFAX: 7.0%
Pre-project loading: 102.4%
23. Facility: PAJUUL TC to PORT ROBSON 138.0 kV L
Contingency: TABER 345/138 kV Transformer Bank
Loading: 103.9% (Rating: 420 MVA)
DFAX: 13.8%
Pre-project loading: 101.3%
24. Facility: PAJUUL TC to SAN LORENZO 3302 138.0 kV line
Contingency: TERRY to HOOKDALE 138.0 kV line
Loading: 103.3% (Rating: 299 MVA)
DFAX: 9.2%
Pre-project loading: 100.9%
25. Facility: DUNLAVY to VIADUCT AVENUE 138.0 kV line
Contingency: PORT ROBSON SUBSTATION to HOLOPAW 138.0 kV
Loading: 101.9% (Rating: 206 MVA)
DFAX: 10.3%
Pre-project loading: 97.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HOOKDLE to COLUMBA 138.0 kV #2	Rebuild line with higher-capacity conductor	\$2,870,000	24% (\$681,000)
CALIFORNIA AVE to SIERRA PACIFIC LUMBER 138.0 kV	Rebuild line with higher-capacity conductor	\$10,671,000	12% (\$1,260,000)
FOOTHILL to WAUGH CHAPEL 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,359,000	17% (\$1,067,000)
COLUMBIA to HOOKDALE 138.0 kV Cir. 1	Rebuild line with higher-capacity conductor	\$2,166,000	24% (\$514,000)
GLEN BURNIE to SANDARIO 138.0 kV Line	Rebuild line with higher-capacity conductor	\$4,603,000	22% (\$1,015,000)
CHANNELVIEW to CALIFORNIA AVE 138.0 kV	Rebuild line with higher-capacity conductor	\$1,713,000	12% (\$202,000)
MCCALL SUBSTTAION to FOOTHILL 138.0 kV	Rebuild line with higher-capacity conductor	\$1,446,000	17% (\$243,000)
FOX RIVER to SANDARIO 138.0 kV	Rebuild line with higher-capacity conductor	\$2,571,000	12% (\$304,000)
CHANNELVIEW SUBSTATION to GEORGIA-PACIFIC CROSSETT LLC SUBSTATION 138.0 kV	Rebuild line with higher-capacity conductor	\$1,989,000	12% (\$235,000)
HUNT CLUB to SIERRA PACIFIC LUMBER 138.0 kV L	Rebuild line with higher-capacity conductor	\$10,854,000	22% (\$2,341,000)
PORT ROBSON to HOLOPAW 138.0 kV	Rebuild line with higher-capacity conductor	\$2,379,000	20% (\$483,000)
GEORGIA-PACIFIC CROSSETT LLC SUBSTATION to FOX RIVER 138.0 kV	Rebuild line with higher-capacity conductor	\$1,897,000	12% (\$224,000)
SANDARIO to STILLWATER HYDRO ELECTRIC PROJECT 138.0 kV Line	Rebuild line with higher-capacity conductor	\$1,390,000	22% (\$301,000)
HUNT CLUB to SALAMANCAFRANK ST 138.0 kV	Rebuild line with higher-capacity conductor	\$2,745,000	22% (\$592,000)
STILLWATER HYDRO ELECTRIC PROJECT to MCCALL SUBSTTAION 138.0 kV	Rebuild line with higher-capacity conductor	\$3,342,000	22% (\$723,000)
TERRY to FRENCH GULCH 138.0 kV line	Rebuild line with higher-capacity conductor	\$3,864,000	20% (\$769,000)
RIO BLANCO 2702 to ALLIANE 138.0 kV	Rebuild line with higher-capacity conductor	\$1,232,000	20% (\$248,000)
BRUCE CROSSING to RIO BLANCO 2702 138.0 kV	Rebuild line with higher-capacity conductor	\$1,937,000	20% (\$390,000)
BRUCE CROSSING to HOLOPAW 138.0 kV	Reconductor line	\$1,365,000	20% (\$275,000)
RALPH to FRENCH GUCH SUBSTATION 138.0 kV	Reconductor line	\$791,000	20% (\$157,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
COLUMBIA to SAN LORENZO 3302 138.0 kV	Reconductor line	\$1,644,000	23% (\$377,000)
TERRY to HOOKDALE 138.0 kV	Reconductor line	\$613,000	20% (\$122,000)
PAJUILL TC to PORT ROBSON 138.0 kV L	Reconductor line	\$704,000	20% (\$140,000)
PAJUILL TC to SAN LORENZO 3302 138.0 kV line	Reconductor line	\$576,000	20% (\$114,000)
VIADUCT AVENUE to DUNLAVY 138.0 kV	Reconductor line	\$498,000	20% (\$100,000)
	Total allocated to project		\$12,877,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.